

Suggested Solutions

Qn	Suggested Solutions	Mark Scheme
1(a)	VisitorName uses a <u>presence check</u>. PartySize uses a <u>type check</u>. PassCode uses a <u>format check</u>.	1 1 1
(b)	Record 1 fails the <u>type check</u> because <code>four</code> is not an integer. Record 2 fails the <u>format check</u> because it does not have <code>v</code> followed by exactly three digits.	1 1
(c)	The value 5 has the <u>required integer data type</u>, so it satisfies the stated validation rule. Validation checks whether data follows specified rules; it <u>does not</u> establish that the entered value matches the real party size.	1 1

Qn	Suggested Solutions	Mark Scheme
2(a)	Groups the bits into the two nibbles <code>1010</code> and <code>1110</code>. Converts these to hexadecimal to give <u>__AE__</u>.	1 1
(b)	Shows the place-value calculation $3 \times 16 + 13$. Gives the <u>correct denary value</u> 61.	1 1
(c)	Decodes <code>01010010</code> as <u>__R__</u>. Decodes <code>01000110</code> as <u>__F__</u>, giving the label RF.	1 1
(d)	The label requires <u>18 bytes</u> .	1
(e)	The same bit pattern must be mapped to the <u>same character at both computers</u> ; otherwise, the receiver may interpret the transmitted characters incorrectly.	1

Qn	Suggested Solutions	Mark Scheme
3(a)	The network within one branch is a <u>__LAN__</u>. The network connecting branches in different locations is a <u>__WAN__</u>. The staff-only private catalogue network is an <u>__intranet__</u>.	1 1 1
(b)	The switch <u>registers or learns the tablet's MAC address</u> when the tablet attempts to join the network. The router <u>assigns the tablet an IP address and stores the allocation in its DHCP table</u>. <i>Do not require DHCP message names, lease negotiation, configuration parameters or NAT.</i>	1 1
(c)	The tablet client <u>sends a catalogue-search request</u> to the server. The server processes the request and <u>returns the matching catalogue data or search result</u> to the client.	1 1

(d)	A protocol provides an <u>agreed set of rules</u> , such as how the request and response are formatted or transmitted. These shared rules allow the client and server to <u>interpret the exchanged data consistently and communicate successfully</u> .	1 1
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Qn	Suggested Solutions	Mark Scheme
4(a)	The comparisons begin with 417, 203 and 586 <u>in this order</u> . The search then compares 144 followed by 392 <u>and stops</u> . The returned index is <u>4</u> .	1 1 1
(b)	The <u>first three selected values</u> are 144, 203, 251. The <u>next two selected values</u> are 392, 417. The <u>completed merged list</u> is [144, 203, 251, 392, 417, 586].	1 1 1
(c)	The <u>first split is correct</u> : [38, 12, 45] and [7, 26, 19], and both sides are split repeatedly until all sublists contain one value. The <u>intermediate merges are correct</u> , including [12, 38], [12, 38, 45], [7, 26] and [7, 19, 26]. The <u>final merge</u> gives [7, 12, 19, 26, 38, 45]. <i>Accept an equivalent diagram that shows every splitting and merging stage and follows the stated rule for an odd-sized list.</i>	1 1 1
(d)	The <u>worst-case time complexity</u> is $O(n \log n)$.	1

Qn	Suggested Solutions	Mark Scheme
5(a)	Studio 2 is stored in the field <u>__Venue__</u> . The <u>complete record</u> is E1044, Chen, C15, Robotics Basics, Lab 3.	1 1
(b)	The course title and venue for C08 are <u>stored repeatedly</u> , causing data redundancy. If one copy is changed without changing the others, the same course may have <u>conflicting titles or venues</u> , causing data inconsistency.	1 1
(c)	EnrolmentID is a suitable candidate key because its value <u>uniquely identifies each record</u> . ParticipantName is not suitable because <u>the same participant name can occur in more than one record</u> , as shown by Aisha.	1 1